

A Look at Key CNN Architectures

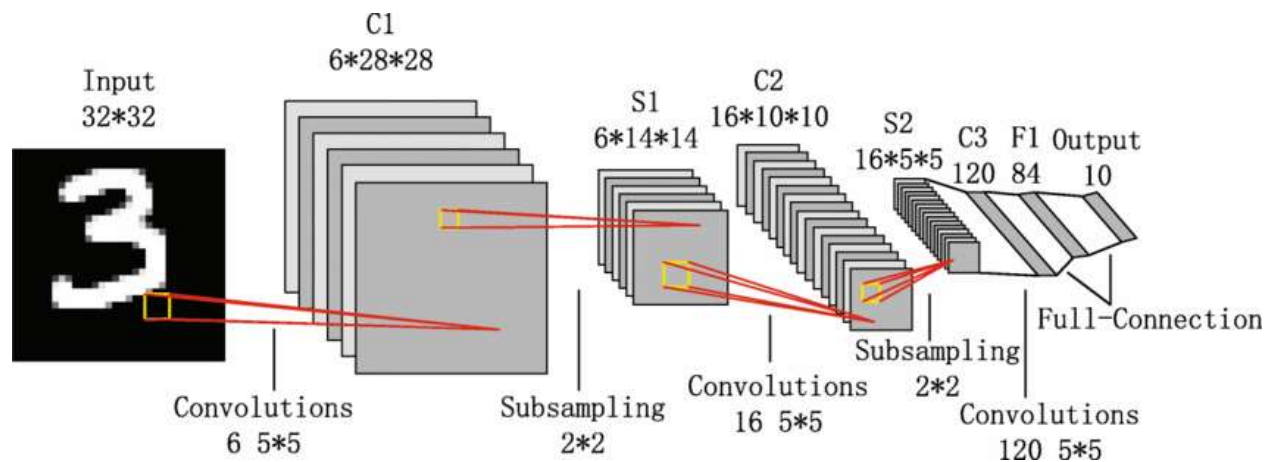
The field of computer vision has been revolutionized by Convolutional Neural Networks (CNNs). Here is a breakdown of several foundational and influential CNN architectures, with a focus on their unique contributions.

LeNet

LeNet was one of the first and most pioneering CNN architectures, introduced by Yann LeCun in the late 1990s. Its primary application was recognizing handwritten digits. While it seems simple today, it established the core building blocks that are still used in modern CNNs.

Key Contributions:

- **Convolutional Layers:** LeNet was one of the first to successfully use convolutional layers to automatically extract features from images. This was a paradigm shift from traditional methods that required manual feature engineering.
- **Pooling Layers:** It introduced pooling (subsampling) layers to reduce the spatial dimensions of the feature maps, which decreased computational complexity and made the model more robust to minor shifts and distortions in the input image.
- **End-to-End Learning:** It demonstrated that a complete system, from raw pixel input to final classification, could be trained as a single network.



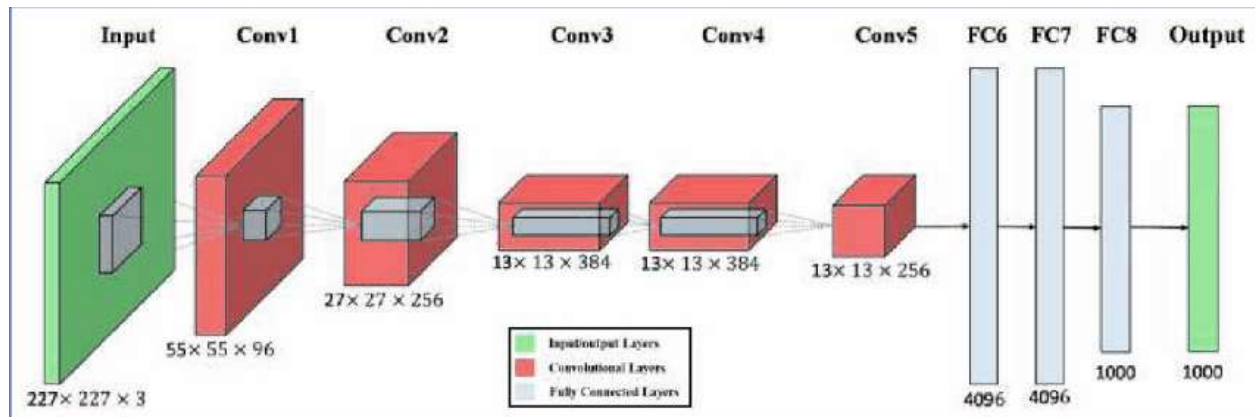
AlexNet

AlexNet burst onto the scene in 2012, winning the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) by a significant margin. Its success is widely considered the beginning of the deep learning revolution in computer vision.

Key Contributions:

- **Deep Architecture:** AlexNet was significantly deeper than previous networks, with 8 layers. This showed that stacking more layers could lead to much better performance.
- **ReLU Activation Function:** It was one of the first networks to use the **Rectified Linear Unit (ReLU)** as its activation function. ReLU helped to solve the **vanishing gradient problem**, allowing for more stable and faster training of deeper networks.

- **Dropout Regularization:** It introduced dropout to prevent overfitting, a common problem in deep neural networks with many parameters.
- **GPU Training:** The model was so large that it required two GPUs to train, highlighting the need for parallel processing in deep learning.

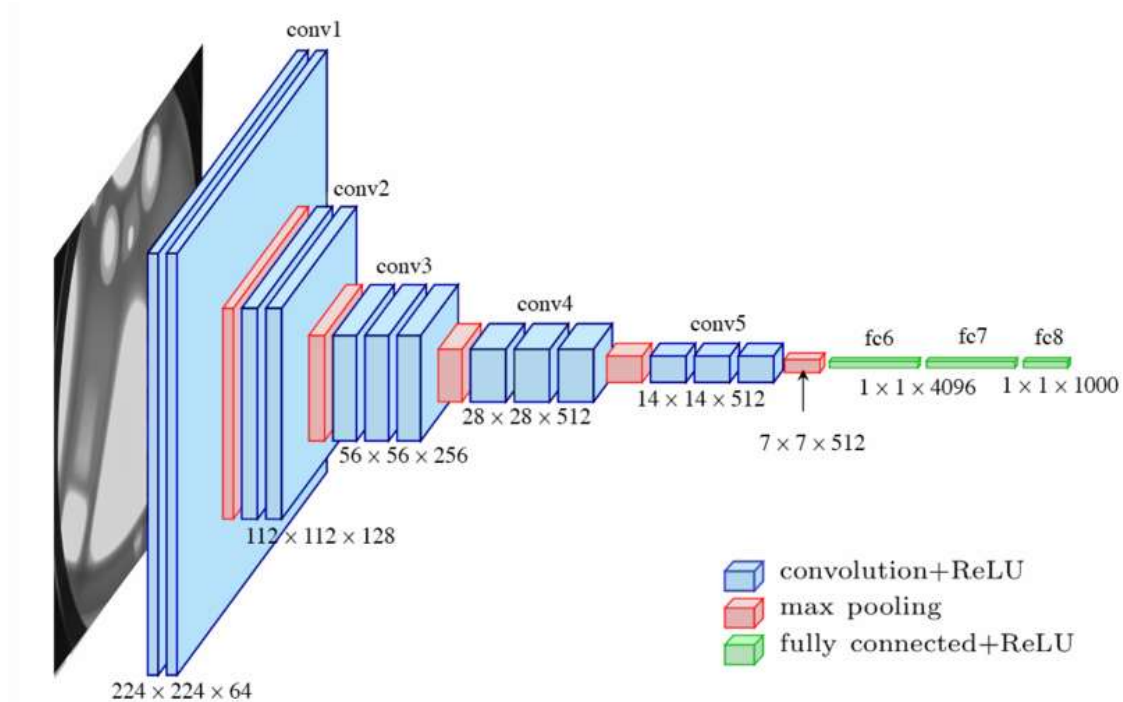


VGG

VGG (Visual Geometry Group) was introduced in 2014 and is famous for its simplicity and depth. Its main goal was to explore the effect of network depth on accuracy in a large-scale image recognition setting.

Key Contributions:

- **Uniform Architecture:** VGG used a very simple and uniform architecture, consisting of a stack of small 3x3 convolutional filters. This showed that depth, rather than a variety of filter sizes, was a crucial component for high performance.
- **Increased Depth:** By using many small filters, VGG was able to build very deep networks (up to 19 layers) with a manageable number of parameters. This demonstrated that deep, uniform architectures could be highly effective.

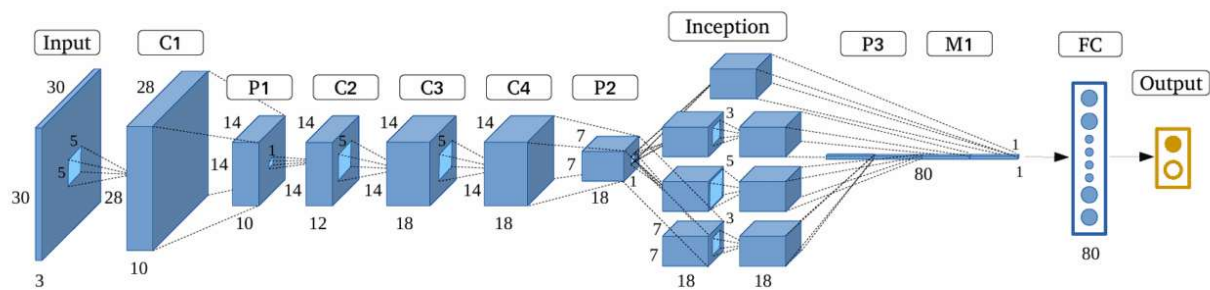


GoogleNet

GoogleNet (also known as Inception) was a major winner of the ILSVRC 2014 competition. It was designed to be both deep and computationally efficient.

Key Contributions:

- **Inception Module:** Its most important contribution was the **Inception module**, which allowed the network to learn multiple feature scales and representations simultaneously. It used 1×1 , 3×3 , and 5×5 convolutions along with a pooling layer in a single block.
- **Global Average Pooling:** It replaced the fully connected layers at the end with a global average pooling layer, which drastically reduced the number of parameters and prevented overfitting.
- **Computational Efficiency:** The use of 1×1 convolutions (pointwise) for dimensionality reduction was a key trick to making the Inception module computationally cheaper, allowing the network to be much deeper and wider than its predecessors.

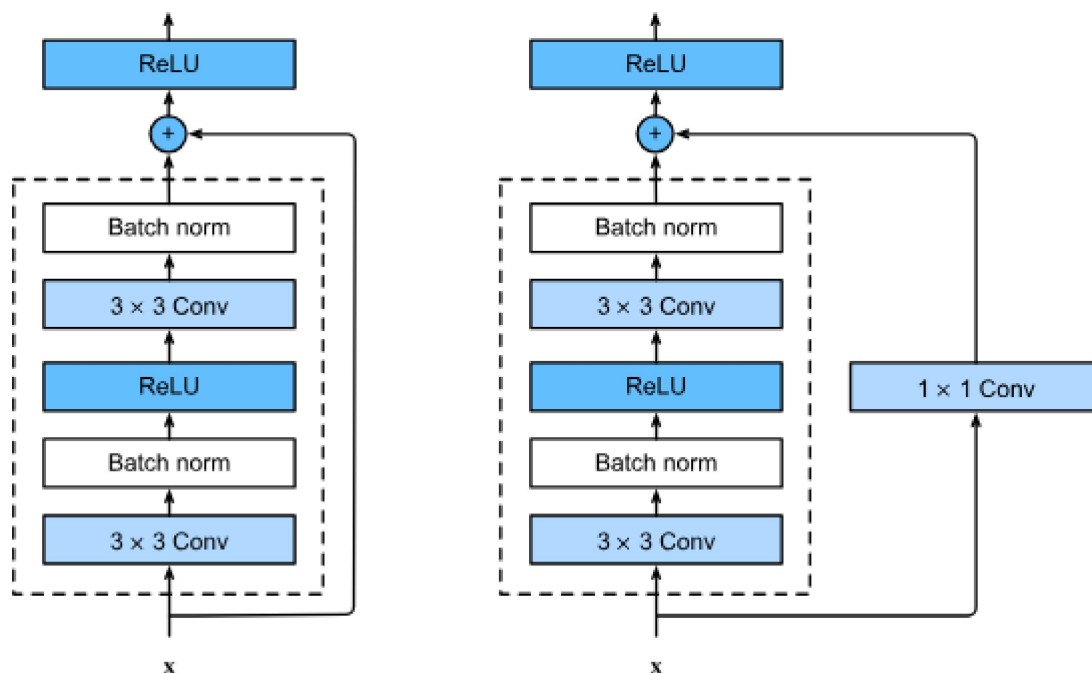


ResNet

ResNet (Residual Network) won the ILSVRC 2015 competition and introduced a fundamental concept that allowed for the training of extremely deep neural networks (up to 152 layers).

Key Contributions:

- **Residual Connections:** This is the most significant contribution of ResNet. The network uses "skip connections" or **residual connections** that allow the input of a layer to be added to its output. This helps to solve the **vanishing gradient problem** and allows the network to be much, much deeper without a significant performance drop.
- **Identity Mapping:** The skip connections essentially enable the network to learn an identity mapping, which is a crucial capability for deep networks. This means that if a layer finds that its output isn't a good fit, it can simply skip over itself and pass its input to the next layer.

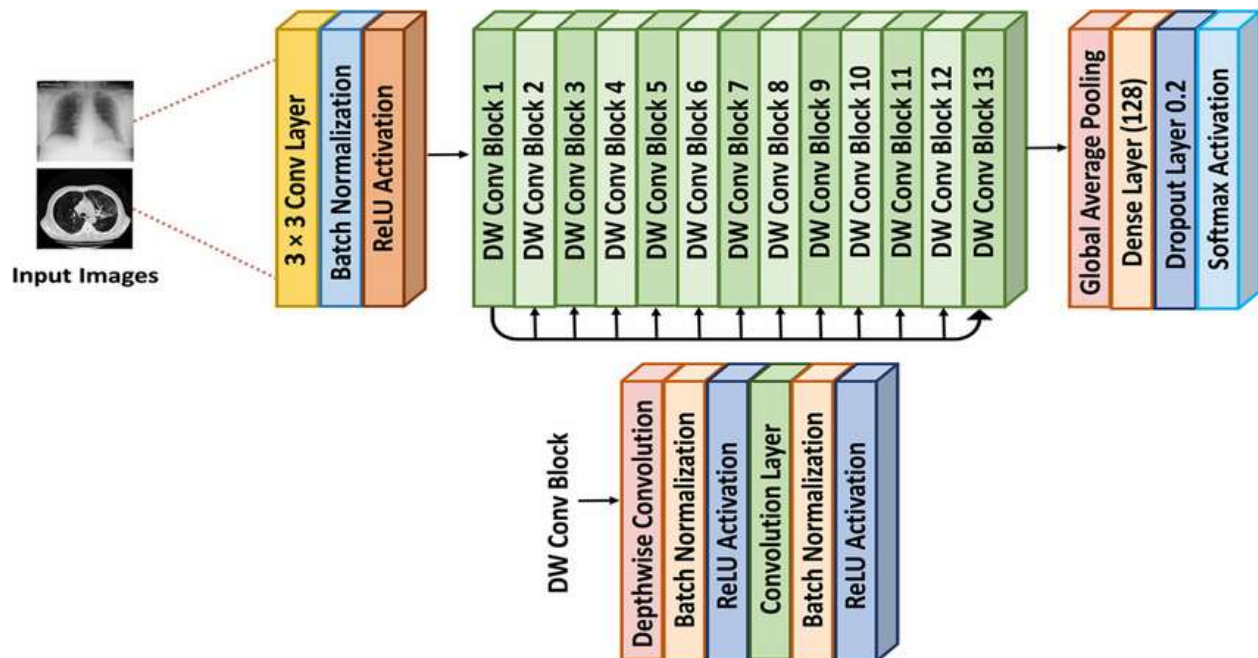


MobileNet

MobileNet is a class of models designed specifically for mobile and embedded vision applications where computational resources and power are limited. 2017

Key Contributions:

- **Depthwise Separable Convolutions:** The core idea behind MobileNet is to replace a standard convolution with a depthwise separable convolution. This breaks a standard convolution into two smaller, more efficient operations: a depthwise convolution (applying a single filter to each input channel) and a pointwise convolution (a 1×1 convolution).
- **Computational Efficiency:** This technique significantly reduces the number of parameters and computations, making the model much faster and smaller while maintaining a high level of accuracy.



EfficientNet

EfficientNet is a recent architecture that systematically explores how to best scale a CNN. It introduces a new scaling method that uniformly scales the network's depth, width, and resolution. 2019

Key Contributions:

- **Compound Scaling:** EfficientNet's main contribution is its **compound scaling** method. It uses a compound coefficient to scale the network's depth, width, and image resolution simultaneously and in a principled way.
- **Optimized Performance:** This uniform scaling approach leads to a family of models that are much more computationally efficient and perform better than previous models, achieving state-of-the-art accuracy with significantly fewer parameters.

These architectures represent major milestones in the development of deep learning for computer vision, each contributing a novel idea that paved the way for the next generation of

models.

